

# Locating and hardening the operating point of intensity-difference squeezing from four-wave mixing in hot $^{85}\text{Rb}$ : a simulation case study

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## Abstract

We report a systematic numerical study of twin-beam intensity-difference squeezing (IDS) generated by double- $\Lambda$  four-wave mixing (FWM) on the  $^{85}\text{Rb}$  D1 line, carried out with an internal steady-state Floquet–Maxwell–Bloch simulation whose full source code is publicly available. Anchoring all fixed parameters to a recent single-laser FWM experiment [Sim *et al.*, Sci. Rep. **15**, 7727 (2025)], we scan the squeezing over the pump one-photon detuning  $\Delta$  and the cell temperature  $T$ , with the two-photon detuning  $\delta$  optimized at each point. The final model predicts a recommended operating point at  $\Delta = -1.50$  GHz (relative to  $F=2 \rightarrow F'=3$ ),  $T = 110^\circ\text{C}$ ,  $\delta = -280$  MHz, where the simulated squeezing reaches  $\xi = -8.10$  dB at a realistic detection efficiency  $\eta = 0.869$  — within 0.7 dB of the detection floor  $10 \log_{10}(1 - \eta) = -8.84$  dB, and 0.32 dB deeper than the best blue-detuned operating point near the conventional experimental choice. A five-parameter tolerance analysis around this point yields, for a +0.5 dB degradation budget, allowed drifts of +2.1 percentage points in detection-path loss,  $-9.6/+4.2$  MHz in two-photon detuning,  $-32/+155$  MHz in one-photon detuning, and  $-3.3/+12^\circ\text{C}$  in cell temperature, with no measurable sensitivity to seed power between 1 and  $64 \mu\text{W}$ ; the resulting troubleshooting priority is loss  $\rightarrow$  two-photon detuning  $\rightarrow$  one-photon detuning  $\rightarrow$  temperature  $\rightarrow$  seed power.

We deliberately present the study as a chronological case study, because three successive, superficially plausible results produced along the way turned out to be spurious: a broad “maximum” that was in fact a detection-floor degeneracy, making the naive question “where is the maximum?” ill-posed; a  $-82$  dB prediction traced to a catastrophically canceling noise formula with the wrong high-gain asymptote; and an extreme-gain optimum created by an objective function blind to the absorption that generates the gain. Each defect was caught by an internal-consistency guardrail (a detection-floor bound, a twin-beam gain-gap criterion, cross-evaluation of independent formulas on identical gain data, and a sign audit of the side-band beat), and we distill these guardrails into a checklist for simulation-based predictions of squeezing experiments.

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## 1 Introduction

Non-degenerate four-wave mixing in hot alkali vapor converts two pump photons into a probe (signal) photon and a conjugate photon. Because the photons are created strictly in pairs, the intensities of the two output beams are so strongly correlated that the noise of their difference  $N_s - N_c$  falls below the shot-noise limit, producing *intensity-difference squeezing* that can be observed with direct balanced detection — no local oscillator required. Since the first strong demonstrations in rubidium vapor [1, 2], the double- $\Lambda$  scheme on the  $^{85}\text{Rb}$  D1 line has become a workhorse source of bright twin beams, with measured squeezing of  $-8$  dB and beyond, and it remains under active development, including single-diode-laser implementations [4] and long-term-stabilized operation [5]. It is also well established experimentally that the best squeezing is *not* obtained at the highest gain: absorption and loss compete with the parametric process, so the optimum operating point is set by a trade-off [3].

For an experimentalist setting up such a source, the practical questions are concrete. Where in the plane of pump one-photon detuning  $\Delta$  and cell temperature  $T$  (equivalently, atomic density) is the squeezing deepest? What physically limits the attainable depth? And how tightly must each knob — detunings, temperature, seed power, detection loss — be controlled before the squeezing degrades noticeably? This paper answers these questions with a first-principles steady-state simulation of the double- $\Lambda$  system, calibrated once against the reference experiment of Sim *et al.* [4] and then used to scan the full operating plane. All computations were performed with an internal simulation code; the complete source, the archived scan data, and the scripts that regenerate every figure in this paper are publicly available on GitHub [6].

We have chosen to present the study *chronologically*, as a case study in five stages, rather than presenting only the final model. The reason is methodological honesty: along the way the simulation produced three successive answers that looked plausible and were later shown to be

wrong — one conceptually ill-posed question, one outright formula bug amplified by floating-point cancellation, and one artifact of an objective function that did not charge the physical cost of the gain it rewarded. Each wrong answer survived until a specific internal-consistency check exposed it. We believe the sequence of failures and the guardrails that caught them are at least as useful to the community as the final numbers, because the same failure modes threaten any simulation-based optimization of a squeezing experiment. Readers interested only in the final predictions can skip to Secs. 7–8; the guardrail checklist is collected in Sec. 9.

## 2 The simulation

### 2.1 Physical model

The atomic system is the standard double- $\Lambda$  configuration on the  $^{85}\text{Rb}$  D1 line: the two hyperfine ground states  $F = 2, 3$  (splitting  $\nu_{\text{HF}} = 3.035732439$  GHz) and the two excited states  $F' = 2, 3$ . A strong pump of power  $P_{\text{pump}}$  drives both legs; a weak seed (probe) is injected on the  $(-)$  Raman branch and a conjugate builds up on the opposite sideband. Rabi frequencies follow from the beam intensities,  $I = 2P/(\pi w_0^2)$  and  $\Omega = \Gamma \sqrt{I/(2I_{\text{sat}})}$ , with the measured beam waists.

For each probe frequency and each atomic velocity class the simulation solves the steady state of the driven system in a Floquet sideband expansion and extracts the dimensionless susceptibility matrix  $(\bar{\chi}_{ss}, \bar{\chi}_{sc}, \bar{\chi}_{cs}, \bar{\chi}_{cc})$  that couples the probe and conjugate fields. After Doppler averaging over the Maxwell velocity distribution, the two sidebands are propagated through the cell with the linearized coupled-mode equation

$$\frac{d}{dz} \begin{pmatrix} \Omega_s \\ \Omega_c^* \end{pmatrix} = M \begin{pmatrix} \Omega_s \\ \Omega_c^* \end{pmatrix}, \quad G_s = |\mathcal{T}_{00}|^2, \quad G_c = |\mathcal{T}_{10}|^2, \quad (1)$$

where  $\mathcal{T} = \exp(ML)$  is the transfer matrix of the cell of length  $L$  and  $G_s, G_c$  are the probe and conjugate power gains. The propagation is segmented along the cell, with the pump power depleted self-consistently segment by segment (energy conservation caps the total gain at  $G_{\text{cap}} = 1 + P_{\text{pump}}/(2P_{\text{seed}})$ ), and includes an iteratively converged longitudinal phase mismatch  $\Delta k_z$ , the Gaussian overlap of the pump and probe modes at their crossing angle  $\theta$ , and Clebsch–Gordan-weighted effective couplings that fold in the Zeeman substructure (a full Zeeman-resolved Floquet solve is deferred). The atomic density follows from the cell temperature through the vapor-pressure curve. Density- and temperature-dependent collisional decoherence is included both for the optical coherences (resonant self-broadening,  $\Gamma_{\text{eff}} = \Gamma + \beta N$ ) and for the ground-state Raman coherence (transit-time floor plus Rb–Rb collisions).

### 2.2 From gains to measured squeezing

For a lossless twin-beam source the photon-number difference is a conserved quantity of the parametric interaction, which fixes the intensity-difference noise (normalized so that  $S = 1$  is the shot-noise limit) to

$$S_{\text{src}} = \frac{(G_s - G_c)^2}{G_s + G_c} \xrightarrow{G_c = G_s - 1} \frac{1}{2G_s - 1}. \quad (2)$$

The high-gain behavior  $S \propto 1/(2G)$  matches the standard result of the experimental literature [4]; an earlier, incorrect version of this formula plays a central role in Sec. 5.

Losses are applied as beam splitters in three tiers. Inside the cell each sideband is attenuated by its own linear absorption: the probe and the conjugate sit at different optical frequencies ( $f_c = 2\Delta - f_{\text{probe}}$  in the rotating frame of the pump) and each samples the  $F=2 \rightarrow F'$  absorption manifold at its own frequency, giving arm transmissions  $\tau_{s,c} = e^{-\text{OD}_{s,c}}$ . The parametric susceptibilities of the four-level model do not contain this linear absorption of the full hyperfine manifold, so it is computed from a separately validated absorption model (verified to  $< 0.1\%$  against laboratory optical-depth data) and imposed on each arm without any adjustable coefficient. A small

amount of pump light scattered into the detection modes is modeled as an incoherent noise contribution anchored to the (validated) pump optical depth,  $\propto \kappa(1 - e^{-\text{OD}_{\text{pump}}})$ , with  $\kappa = 0.1$  the only residual coefficient of this *extended excess-noise model*; the results below are insensitive to  $\kappa$  (the artifact suppression reported in Stage IV persists even at  $\kappa = 0$ ). Unequal arm losses are combined in a standard balanced-detector noise model for two partially correlated beams. Finally, the total detection efficiency  $\eta = \text{QE}(1 - \ell)$ , with  $\ell$  the post-cell optical loss, acts symmetrically on both arms:

$$S(\eta) = \eta S_{\text{cell}} + (1 - \eta), \quad \xi = 10 \log_{10} S(\eta) \text{ [dB]}. \quad (3)$$

Because  $S_{\text{cell}} \geq 0$ , Eq. (3) implies the *detection floor*

$$\boxed{\xi \geq \xi_{\text{floor}} = 10 \log_{10}(1 - \eta)} \quad (4)$$

for every operating point: no simulated (or measured) squeezing can be deeper than the vacuum noise admitted by the detection loss. Equation (4) requires only  $S_{\text{cell}} \geq 0$  and is therefore independent of any modeling error upstream — which is precisely what makes it useful as a guardrail (Sec. 3).

Two properties of this chain are worth noting. First,  $\eta$  is pure post-processing: the gains and in-cell losses do not depend on it, so a single scan of the  $(\Delta, T)$  grid can be re-read at any detection efficiency essentially for free. Second, loss can only move  $S$  *toward* shot noise; it can never push the difference noise above it. Observed anti-squeezing (positive dB) therefore always signals excess noise — pump-noise coupling, spontaneous Raman scattering, or leakage of the large anti-squeezed intensity-*sum* noise through detector imbalance — and never mere attenuation.

### 2.3 Conventions

Throughout the paper, the one-photon detuning  $\Delta$  is the pump detuning from the  $^{85}\text{Rb}$  D1  $F=2 \rightarrow F'=3$  transition,  $\omega_{\text{pump}} = \omega(F=2 \rightarrow F'=3) + \Delta$ , matching the convention of Ref. [4]. The two-photon detuning  $\delta$  is defined through the seed frequency on the (−) Raman branch,

$$\omega_{\text{seed}} = \omega_{\text{pump}} - \nu_{\text{HF}} + \delta, \quad (5)$$

so that the physical pump–seed beat note is  $\nu_{\text{HF}} - \delta$  (the sign convention in Eq. (5) is the subject of the audit in Sec. 7). The pump–probe crossing angle is  $\theta = 0.32^\circ$ , the geometry of the reference experiment, unless stated otherwise. The measured squeezing  $\xi$  is always reported in dB, with negative values meaning noise below the shot-noise limit.

### 2.4 Calibration and reference parameters

All parameters other than the scanned variables are pinned to the reference experiment [4] and are listed in Table 1. The line-strength scale of the propagation is fixed by physical coupling normalization plus one residual calibration factor anchored at the reference operating point; this factor was *not* re-tuned after any of the model revisions described below. With  $\text{QE} = 0.92$  and post-cell loss  $\ell = 5.5\%$ , the reference detection efficiency is  $\eta = 0.8694$  and the detection floor is  $-8.8406$  dB.

### 2.5 The five stages at a glance

Table 2 summarizes how the model and the conclusions evolved. Each stage is described in its own section below. Only the Stage-V numbers are quantitatively current; every earlier coordinate or depth quoted in this paper is explicitly historical and is accompanied by the reason it was superseded.

**Table 1:** Reference parameters (fixed in all scans unless stated otherwise).

| Quantity                                 | Value                   | Note                     |
|------------------------------------------|-------------------------|--------------------------|
| Pump power $P_{\text{pump}}$             | 600 mW                  | Ref. [4]                 |
| Seed power $P_{\text{seed}}$             | $8 \mu\text{W}$         | Ref. [4]                 |
| Cell length $L$                          | 12.5 mm                 |                          |
| Pump / seed waists                       | 530 / 330 $\mu\text{m}$ |                          |
| Pump–probe angle $\theta$                | $0.32^\circ$            | default geometry         |
| Raman branch                             | (–)                     | standard seeded readout  |
| Detector quantum efficiency              | 0.92                    | recalibrated in Stage II |
| Post-cell optical loss $\ell$            | 5.5%                    |                          |
| $\eta = \text{QE}(1 - \ell)$             | 0.8694                  |                          |
| Detection floor $10 \log_{10}(1 - \eta)$ | –8.8406 dB              | Eq. (4)                  |

**Table 2:** Chronology of the study. Each stage’s scan was run with the model content accumulated up to that row.

| Stage    | Model content added                                                                                                                                                       | Outcome / defect exposed                                                                                                                   |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| I (§3)   | lossless parametric source + detection efficiency                                                                                                                         | “maximum squeezing” is ill-posed: everything collapses onto the detection floor; objective reframed as an <i>efficient-frontier</i> search |
| II (§4)  | full propagation: in-cell probe absorption, segmented pump depletion, phase mismatch, Gaussian overlap, CG-weighted Zeeman corrections; detection efficiency recalibrated | in-cell loss creates a genuine interior optimum in $T$ ; transparency window identified                                                    |
| III (§5) | wide ideal-detection scan; then corrected difference-noise formula, collisional decoherence, twin-beam gap gate                                                           | spurious –82 dB traced to a formula bug (catastrophic cancellation, wrong asymptote)                                                       |
| IV (§6)  | per-sideband linear absorption + pump scatter (“extended excess-noise model”)                                                                                             | extreme-gain artifact removed ( $G_s 636 \rightarrow 27$ ); red/blue lobe near-degeneracy identified                                       |
| V (§7)   | sideband-beat sign correction                                                                                                                                             | optimum relocates to (–1.50 GHz, 110 °C); final frontier, ideal limit, geometry study, tolerance analysis                                  |

### 3 Stage I: the ill-posed maximum

The study began with the most obvious question: fix everything at the reference values and find the  $(\Delta, T)$  that gives the deepest squeezing. The first scan (684 grid points,  $\Delta \in [-0.5, 3.0]$  GHz,  $T \in [60, 150]$  °C, with  $\delta$  optimized at each point) used the simplest defensible model: a lossless parametric source characterized by  $(G_s, G_c)$ , followed by the detection efficiency of Eq. (3). At this stage the efficiency was calibrated at  $\eta = 0.855$ , giving a floor of –8.385 dB. (The detector quantum efficiency was recalibrated from 0.905 to 0.92 in Stage II, moving the floor to –8.841 dB; the Stage-I numbers quoted here are historical.)

The result was unambiguous and, at first sight, disappointing: the deepest squeezing found anywhere on the grid was –8.385 dB — the detection floor itself, reproduced to four decimal places — and *no* grid point fell below the floor (0/684). More than a quarter of the grid sat within 0.01 dB of the floor. The “maximum” was not a point but a broad plateau (Fig. 2).

The mechanism is elementary and worth stating precisely, because it recurs throughout the paper (Fig. 1). For balanced twin beams the source noise falls off with gain — in the corrected formula of Eq. (2) as  $S_{\text{src}} \simeq 1/(2G_s)$  — so beyond a modest “knee” ( $G_s \gtrsim 30$  at the reference

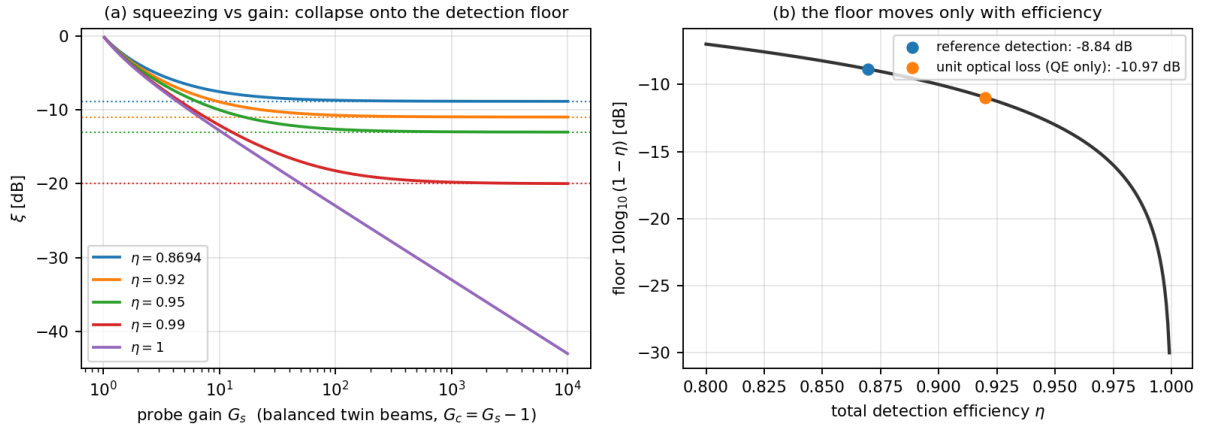
efficiency) the term  $\eta S_{\text{src}}$  becomes negligible compared with the vacuum leakage  $(1 - \eta)$ , and

$$S(\eta) \longrightarrow 1 - \eta. \quad (6)$$

Gain is exponential in atomic density and hence double-exponential in temperature, so the knee is crossed almost everywhere at high temperature, and a large two-dimensional region maps onto one and the same number. Two consequences follow:

1. **The question “where is the maximum?” is ill-posed** at finite detection efficiency. The answer is a degenerate plateau whose height is a property of the *detector*, not of the atoms. The only meaningful optimization is the *efficient frontier*: the operating point that reaches the floor at the lowest cost in temperature, gain, and optical depth — quantities that also control the excess-noise channels (pump scatter, collisional dephasing, anti-squeezed sum-noise leakage) that a static model does not capture.
2. **The floor is a guardrail.** Equation (4) holds for *any* physical source, however elaborate the propagation model. Any simulated point deeper than  $10 \log_{10}(1 - \eta)$  is a numerical or coding error by construction. This check is applied to every scan in this paper; its underlying spirit — distrust any implausibly deep value — is also what eventually exposed the Stage-III bug, in a scan where the floor itself had been removed.

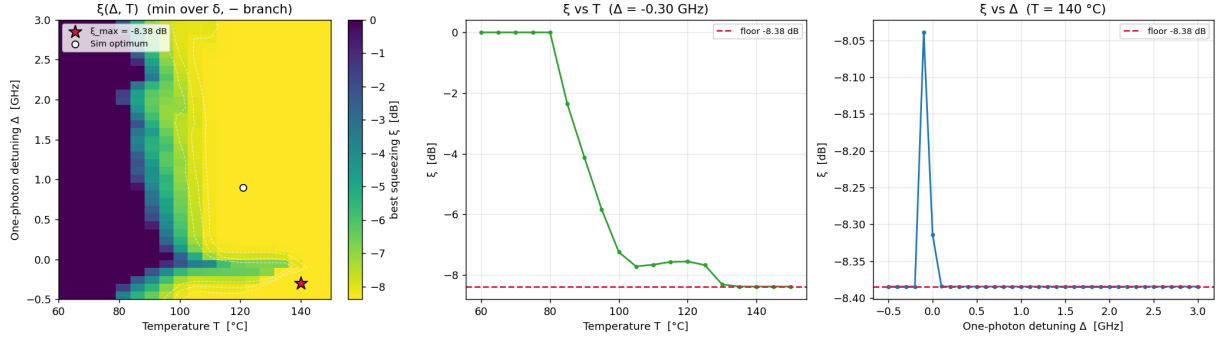
The gain lever and the efficiency lever separate cleanly: at fixed  $\eta = 0.855$ , raising  $G_s$  from 10 to 300 improves  $\xi$  by less than 0.2 dB, whereas at fixed high gain, raising  $\eta$  from 0.855 to 0.95 deepens the floor from  $-8.4$  to  $-13.0$  dB. In this class of experiments the squeezing depth is bought at the detector, not in the vapor cell.



**Figure 1:** The detection-floor mechanism (analytic, using Eqs. (2)–(3) with balanced twin beams). (a) Squeezing versus probe gain for several detection efficiencies: each finite- $\eta$  curve saturates at its floor  $10 \log_{10}(1 - \eta)$  (dotted), and only  $\eta = 1$  improves indefinitely. The marginal value of gain is essentially exhausted by  $G_s \approx 30$ . (b) The floor as a function of  $\eta$ ; the two markers correspond to the recalibrated reference detection ( $\eta = 0.8694$ ) and to a hypothetical lossless setup limited only by the detector quantum efficiency ( $\eta = 0.92$ ).

## 4 Stage II: in-cell loss creates a real interior optimum

The Stage-I model treated the vapor as a pure gain medium. In reality the twin beams propagate through an optically thick resonant vapor, and the absorption they suffer *inside* the cell is not part of the detection efficiency — it depends on  $(\Delta, T)$  and grows steeply with density. Stage II therefore upgraded the propagation to the full model of Sec. 2 (segmented pump depletion, self-consistent phase mismatch, Gaussian overlap, CG-weighted Zeeman corrections) and, crucially, added the in-cell optical depth of the probe as a distributed loss,  $S_{\text{cell}} = \tau S_{\text{src}} + (1 - \tau)$  with



**Figure 2:** Archived Stage-I scan (shown as produced at that stage; the original scan archive was later overwritten, so the figure is reproduced from the archived image). Left:  $\xi(\Delta, T)$  with the low-temperature no-gain region ( $\xi = 0$ ) giving way to the floor plateau at high temperature. Middle: temperature dependence at the optimal detuning — monotonic asymptotic approach to the floor. Right: detuning dependence at fixed  $T$ . Computed with the Stage-I model (pre-correction noise formula, no in-cell loss) and the Stage-I efficiency calibration ( $\eta = 0.855$ , floor  $-8.385$  dB); the plateau phenomenology, though not the numbers, survives all later model revisions.

$\tau = e^{-\text{OD}}$ . The detector quantum efficiency was simultaneously recalibrated to 0.92 against the reference experiment’s detection budget, setting the floor at  $-8.8406$  dB, its final value.

This changed the phenomenology in one essential way: at detunings where the vapor absorbs,  $\xi(T)$  is no longer monotonic. Low temperature starves the gain; high temperature multiplies the optical depth; and in between a genuine interior optimum appears. A map of the in-cell optical depth shows an absorption wedge opening around  $\Delta \in [-1.2, +0.5]$  GHz for  $T \gtrsim 90^\circ\text{C}$ , flanked by transparency: detunings a few GHz to the red of the  $F=2 \rightarrow F'=3$  line stay optically thin up to high temperature (Fig. 3).

As computed at this stage, the global optimum sat in the transparency window ( $\Delta = -2.2$  GHz,  $T = 140^\circ\text{C}$ ) exactly on the recalibrated floor, and the cheapest operating point reaching within 0.1 dB of it was at  $\Delta \approx -1.9$  GHz,  $T \approx 120^\circ\text{C}$  with a moderate gain  $G_s \approx 32$ ; a fixed readout at the reference experiment’s detuning ( $+0.9$  GHz,  $120^\circ\text{C}$ ) gave  $-7.9$  dB, consistent in magnitude with the measured  $-7.8$  dB of Ref. [4]. These specific coordinates did not survive the later corrections (Secs. 5–7) and are quoted only to document the trajectory of the study. What did survive was the structural conclusion: *in-cell absorption converts the degenerate plateau into a real optimization problem*, with a temperature-detuning trade-off of exactly the kind long emphasized in the experimental literature [3].

## 5 Stage III: a spurious $-82$ dB and the formula bug

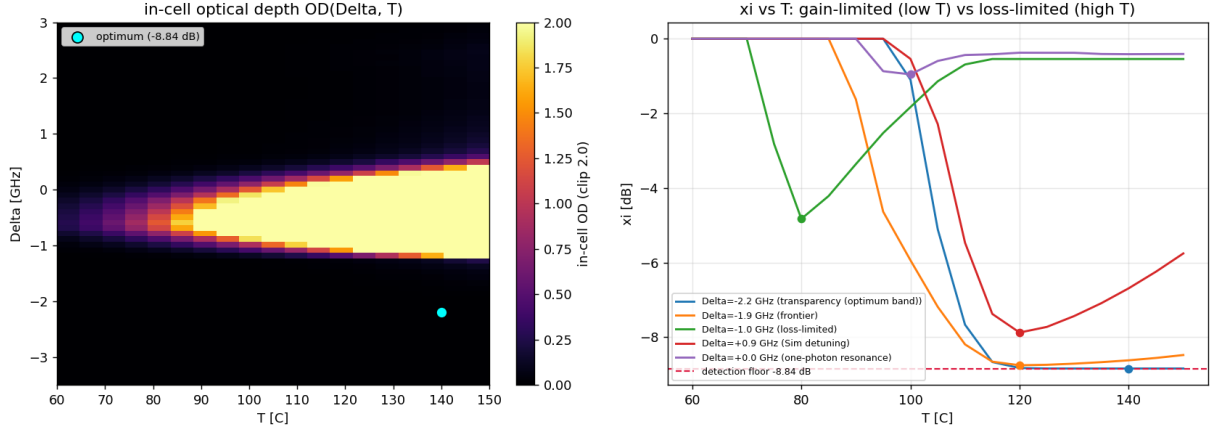
### 5.1 The ideal-detection question and an implausible answer

With realistic detection, the floor hides everything deeper than  $-8.84$  dB. It is nevertheless natural to ask what the *source* itself could deliver: set  $\text{QE} = 1$  and post-cell loss to zero, so the floor recedes to  $-\infty$ , and rescan. Because  $\eta$  is pure post-processing (Sec. 2.2), removing the floor costs nothing on an already-solved grid; for this stage the grid was also widened and refined.

The resulting wide, dense ideal-detection scan ( $\Delta \in [-6, +8]$  GHz,  $T \in [50, 220]^\circ\text{C}$ , 19389 grid points) returned a global optimum of

$$\xi = -81.95 \text{ dB} \quad \text{at} \quad \Delta = -2.10 \text{ GHz}, \quad T = 142.5^\circ\text{C}, \quad G_s \simeq 1.26 \times 10^4. \quad (7)$$

Nothing in the scan diagnostics flagged it: no NaNs appeared, the optimum was interior to the grid, and the value was even consistent with the (then-trusted) high-gain asymptote of the noise formula. A  $-82$  dB source term is not physically forbidden — but it should have prompted the question of whether the formula producing it was trustworthy at  $G \sim 10^4$ .



**Figure 3:** Archived Stage-II diagnostic (shown as produced at that stage; the original scan archive was later overwritten). Left: in-cell optical depth  $OD(\Delta, T)$ , showing the absorption wedge around  $\Delta \in [-1.2, +0.5]$  GHz opening with temperature, and the transparency window to its red side. Right:  $\xi(T)$  at representative detunings — transparent detunings descend monotonically toward the floor, absorbing detunings turn around at an interior optimum. Computed with the Stage-II model (pre-correction noise formula, probe-arm loss only); the wedge geometry and the interior-optimum mechanism persist in all later stages.

## 5.2 The diagnosis: two formulas on identical gain data

The intensity-difference noise formula in use at that time was

$$S_{\text{src}}^{(\text{deprecated})} = \frac{G_s + G_c - 2\sqrt{G_s G_c - 1}}{G_s + G_c}, \quad (8)$$

which has the high-gain asymptote  $5/(8G_s^2)$  for balanced beams. The correct expression follows from the conservation of the photon-number difference in lossless parametric gain and is Eq. (2),  $S_{\text{src}} = (G_s - G_c)^2/(G_s + G_c)$ , with asymptote  $1/(2G_s)$  — matching the standard result of the experimental literature [4]. The two expressions agree only at  $G_s = 2$ ; at high gain the deprecated formula overstates the squeezing by tens of dB, and, worse, its numerator is the difference of two nearly equal large numbers, so floating-point cancellation corrupts it precisely in the regime where it is already wrong.

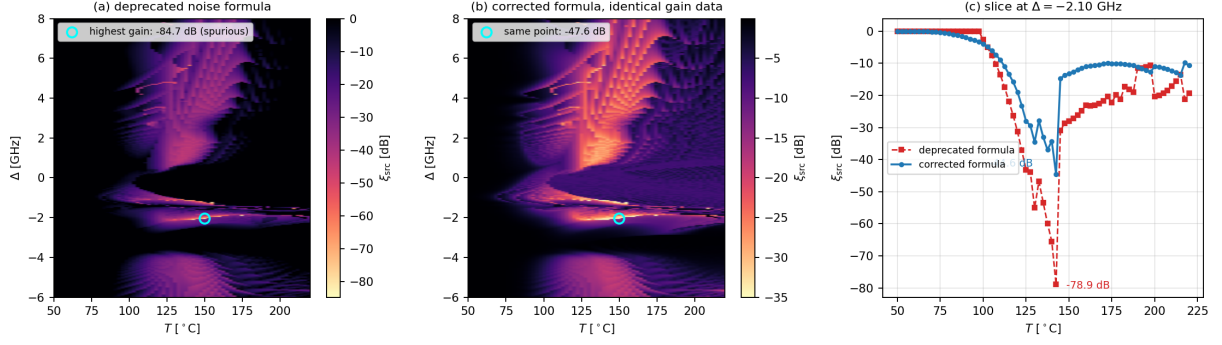
The decisive test required no new physics: *evaluate both formulas on identical archived gain pairs* ( $G_s, G_c$ ) (Fig. 4). At the highest-gain point of the surviving archive ( $\Delta = -2.05$  GHz,  $T = 150^\circ\text{C}$ ,  $G_s = 12828$ ), the deprecated formula gives  $-84.7$  dB — reproducing the magnitude of the spurious optimum — while the corrected formula gives  $-47.6$  dB for the same pair. Along the  $\Delta = -2.10$  GHz slice at  $T = 142.5^\circ\text{C}$  ( $G_s = 6608$ ), the values are  $-78.9$  dB versus  $-44.6$  dB. The spurious result was therefore not exotic physics but a wrong asymptote amplified by catastrophic cancellation.

Re-scanning the same wide grid with the corrected formula moved the ideal source limit to  $\xi = -31.3$  dB at  $\Delta = -2.10$  GHz,  $T = 130^\circ\text{C}$ ,  $G_s \approx 415$  — fifty decibels shallower. (This number, too, was later revised in Stages IV–V; see below.)

## 5.3 Collateral hardening: the gain-gap consistency gate

The corrected formula introduced a new failure mode of its own. Since  $S_{\text{src}} \propto (G_s - G_c)^2$ , any point where the two gains happen to be *equal* reports perfect squeezing — even in background regions with no parametric interaction at all, where  $G_s \approx G_c \approx 1$ . Physically, lossless seeded parametric gain enforces  $G_s - G_c = 1$ ; departures signal either absorption or a numerically untrustworthy point. All scans from this stage onward therefore apply a **twin-beam gain-gap consistency**





**Figure 4:** The Stage-III formula bug, demonstrated on identical archived gain data from the wide ideal-detection scan (source term only; in-cell loss factored out). (a) The deprecated formula, Eq. (8), produces a spurious ultra-deep region; at the archive’s highest-gain point (circle) it yields  $-84.7$  dB, reproducing the magnitude of the originally reported  $-82.0$  dB optimum. (b) The corrected formula, Eq. (2), evaluated on the very same  $(G_s, G_c)$  pairs. (c) Slice at  $\Delta = -2.10$  GHz: the two formulas diverge exactly where the gain grows, by  $\sim 34$  dB at  $T = 142.5$  °C.

**criterion:** a point is trusted only if

$$0.5 \leq G_s - G_c \leq 1.5, \quad (9)$$

and  $\delta$ -optimization is restricted to trusted candidates. At the reference operating point the gap stays within  $[0.55, 0.98]$  across the full  $\delta$  scan, comfortably inside the gate.

At the same stage, the collisional decoherence channels of Sec. 2 were added to the model. With them, the finite-detection squeezing acquires a temperature optimum near  $T \sim 120$ – $135$  °C and degrades at higher temperature — the first time the simulation reproduced the experimentally familiar high-temperature rollover rather than saturating at the floor indefinitely.

The general lesson of Stage III is transferable: when a physics formula is corrected, every downstream consumer must be re-verified, and an incorrect formula may have been silently *masking* out-of-domain evaluations that the corrected formula then exposes (here, the  $G_s = G_c$  background points that necessitated the gate of Eq. (9)).

## 6 Stage IV: gain without cost — the missing absorption channels

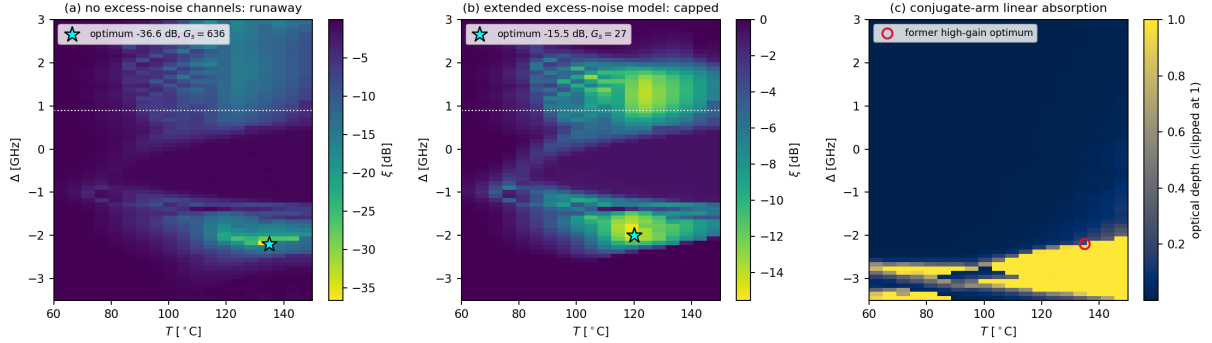
### 6.1 An internally contradictory optimum

After the formula correction, the finite-detection scans still placed the global optimum at  $\Delta = -2.20$  GHz,  $T = 135$  °C with a probe gain of  $G_s = 636$  — while simultaneously reporting an in-cell optical depth of essentially zero at that point. That combination is self-contradictory: a parametric gain of several hundred requires a strong near-resonant atomic response, and a strong response absorbs. The contradiction arose because the objective function was blind to two costs. First, the noise formula Eq. (2) depends only on the gain *difference*: with  $G_s - G_c \approx 1$ , a point with  $G_s = 636$  scores as well as one with  $G_s = 27$ , even though nothing in the model charged for the absorption that generated the factor-600 gain. Second, only the *probe* arm’s absorption was being applied: at far-red detunings the probe sits several GHz from resonance and is indeed transparent, but the *conjugate* (at  $f_c = 2\Delta - f_{\text{probe}}$ ) then lands near the  $F=2 \rightarrow F'$  manifold and is strongly absorbed — at zero model cost.

### 6.2 The extended excess-noise model

Stage IV closed both holes (Sec. 2.2): each sideband is attenuated by the validated linear absorption at its *own* optical frequency (no adjustable parameter), unequal arm losses enter through

the balanced-detector noise model, and pump scatter anchored to the pump optical depth is added with the single residual coefficient  $\kappa$ . The effect on the artifact was decisive: the former optimum at  $(-2.20 \text{ GHz}, 135^\circ\text{C})$ , where the conjugate-arm optical depth is in fact 0.22, dropped from  $-8.83$  to  $-5.41 \text{ dB}$ , and the global optimum moved to a moderate-gain point ( $G_s = 27$  as computed at this stage). Removing the detection floor ( $\eta = 1$ ) confirmed that the change is source physics rather than a detection effect (Fig. 5): without the excess-noise channels the ideal scan ran away to  $-36.6 \text{ dB}$  at the  $G_s = 636$  coordinate, while the extended model capped the ideal optimum at  $-15.5 \text{ dB}$  at the same moderate-gain location as the finite-detection optimum.



**Figure 5:** Stage-IV hardening, shown at ideal detection ( $\eta = 1$ ) where the contrast is not compressed by the detection floor (all panels use Stage-IV-era scans, i.e. the sideband-beat sign error corrected in Stage V is still present, so the coordinates are historical). (a) Without the excess-noise channels the ideal optimum (star) runs away to  $-36.6 \text{ dB}$  at an extreme gain,  $G_s = 636$ , at a nominally transparent point. (b) With per-sideband absorption and pump scatter charged, the ideal optimum is capped at  $-15.5 \text{ dB}$  at a moderate gain,  $G_s = 27$ . (c) The mechanism: the conjugate-arm linear optical depth is large exactly where the uncharged model had found free gain (circle marks the former optimum). The dotted horizontal line marks the reference experiment’s detuning  $+0.9 \text{ GHz}$ .

### 6.3 An honest near-degeneracy

With the artifact removed, a physical question remained: the optimum stayed at *red* pump detuning ( $\Delta \approx -2 \text{ GHz}$  as computed at this stage), whereas the experimental literature operates at  $\Delta \approx +0.8\text{--}1.0 \text{ GHz}$  [1, 2, 4]. The red-side preference is not a numerical accident. At  $\Delta \approx -2 \text{ GHz}$  the pump sits about  $1 \text{ GHz}$  to the blue of the  $F=3 \rightarrow F'=3$  manifold — the mirror image of the conventional operating point, but anchored on the *more populated*  $F = 3$  ground state ( $7/12$  of the population versus  $5/12$  in  $F = 2$ ), and with the seeded probe pushed to a spectrally clean far-red position where its absorption is small. Both effects are real physics. However, the model is a steady-state description with a fixed ground-state population distribution: it does not include the optical pumping by which a pump parked near the  $F = 3$  manifold would deplete  $F = 3$  itself. That depletion is precisely what should erode the red-side advantage in a real experiment. The two lobes are, moreover, nearly degenerate —  $0.28 \text{ dB}$  apart at this stage,  $0.32 \text{ dB}$  in the final model. We therefore report the red/blue asymmetry as a model-level near-degeneracy, not as a firm experimental prediction; resolving it requires optical-pumping dynamics beyond the present model.

## 7 Stage V: the beat-sign audit and the final operating point

### 7.1 The last defect: a sign error in the sideband beat

The Floquet solver takes as input the beat frequency between the pump and the seeded sideband. With the scan coordinate defined by Eq. (5), the physical beat is  $|\omega_{\text{pump}} - \omega_{\text{seed}}| = \nu_{\text{HF}} - \delta$  on the  $(-)$  branch. An audit of the solver input revealed that all previous stages had used  $\nu_{\text{HF}} + \delta$  —

the wrong sign of  $\delta$  for the standard seeded readout. Because  $|\delta|$  is small ( $\lesssim 0.5$  GHz) compared with  $\nu_{\text{HF}}$ , the error had distorted rather than destroyed the landscape, which is why it survived four stages of scrutiny; it was caught only by writing the beat relation down symbolically and checking the code paths against it. After the correction (applied identically in both numerical code paths) the full regression baseline of the simulation was re-anchored, and every scan below was recomputed from scratch.

## 7.2 The final finite-detection frontier

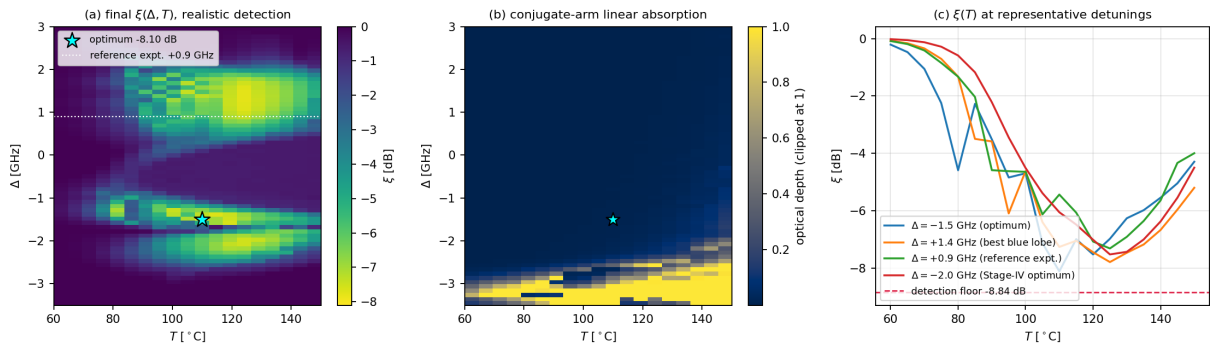
Figure 6 and Table 3 give the corrected finite-detection landscape at the reference efficiency  $\eta = 0.8694$  ( $\Delta \in [-3.5, +3.0]$  GHz,  $T \in [60, 150]$  °C,  $\delta$  optimized within the trust gate at each point). The global optimum moves from the Stage-IV coordinate to

$$\Delta = -1.50 \text{ GHz}, \quad T = 110 \text{ }^\circ\text{C}, \quad \delta = -280 \text{ MHz} : \quad \xi = -8.10 \text{ dB}, \quad G_s = 22.6 \quad (10)$$

with a healthy gain gap  $G_s - G_c = 0.62$  and a small probe optical depth ( $\approx 0.02$ ). The former Stage-IV optimum at  $(-2.0 \text{ GHz}, 120 \text{ }^\circ\text{C})$ , re-evaluated with the corrected beat, is no longer optimal ( $-7.00 \text{ dB}$ ,  $G_s = 7$ ). The best blue-side operating point is  $+1.40 \text{ GHz}$ ,  $125 \text{ }^\circ\text{C}$  with  $-7.78 \text{ dB}$  —  $0.32 \text{ dB}$  shallower than the red-side optimum, preserving the near-degeneracy discussed in Sec. 6. A fixed readout at the reference experiment’s nominal point ( $+0.90 \text{ GHz}$ ,  $121 \text{ }^\circ\text{C}$ ,  $\delta = -8 \text{ MHz}$ ) gives  $-7.20 \text{ dB}$ , to be compared with the measured  $-7.8 \text{ dB}$  and gain  $\approx 15$  of Ref. [4] (simulated  $G_s = 14.0$ ): the simulation reproduces the measured squeezing scale, gain scale, and temperature scale at the experiment’s own operating point.

**Table 3:** Final (Stage-V) finite-detection results at  $\eta = 0.8694$ . The reference-experiment row is a fixed readout at the nominal coordinates of Ref. [4]; all other rows optimize  $\delta$  within the trust gate.

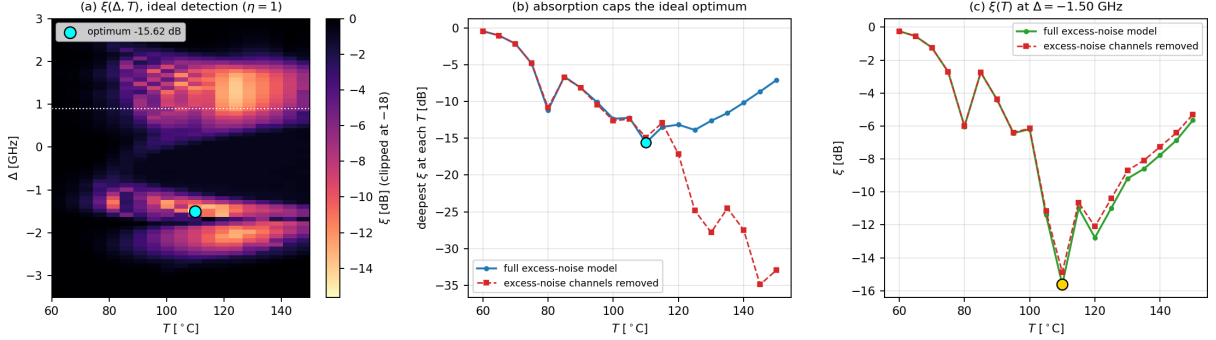
| Point                          | $\Delta$ [GHz] | $T$ [°C] | $\delta$ [MHz] | $\xi$ [dB] | $G_s$ | $G_s - G_c$ |
|--------------------------------|----------------|----------|----------------|------------|-------|-------------|
| Global optimum                 | −1.50          | 110      | −280           | −8.10      | 22.6  | 0.62        |
| Best blue-side lobe            | +1.40          | 125      | 0              | −7.78      | 39.0  | 0.70        |
| Reference-experiment readout   | +0.90          | 121      | −8             | −7.20      | 14.0  | 0.90        |
| Stage-IV optimum, re-evaluated | −2.00          | 120      | −280           | −7.00      | 7.0   | 0.85        |



**Figure 6:** Final finite-detection landscape ( $\eta = 0.8694$ , floor  $-8.84 \text{ dB}$ ). (a)  $\xi(\Delta, T)$  with the global optimum (star) at  $(-1.50 \text{ GHz}, 110 \text{ }^\circ\text{C})$  and the reference experiment’s detuning (dotted). (b) Conjugate-arm linear optical depth: the excess-noise channel that suppresses the far-red high-gain region. (c)  $\xi(T)$  at representative detunings; the red- and blue-side lobes are nearly degenerate, and no curve crosses the detection floor (dashed). The visible jaggedness of the slices reflects the  $17.5 \text{ MHz}$   $\delta$ -grid of the survey scan; the tolerance analysis of Sec. 8 uses a  $5 \text{ MHz}$  grid.

### 7.3 Ideal-detection source limit

Re-reading the same scan at  $\eta = 1$  (Fig. 7) removes the detection floor without moving the optimum: the deepest ideal-detection squeezing is  $-15.62$  dB at exactly the same  $(\Delta, T, \delta)$  as the finite-detection optimum. The floor thus only compresses the landscape; the *location* of the optimum is set by in-cell absorption, i.e. it is source physics. The contrast with the cost-blind objective remains stark even after the beat correction: without the excess-noise channels the ideal scan runs away to  $-34.9$  dB at a  $G_s \approx 1400$  coordinate that the full model evaluates at just  $-3.9$  dB. On the blue side the ideal-detection lobe is  $-13.9$  dB,  $1.7$  dB shallower than the red-side optimum — the near-degeneracy widens when the floor no longer compresses both lobes.



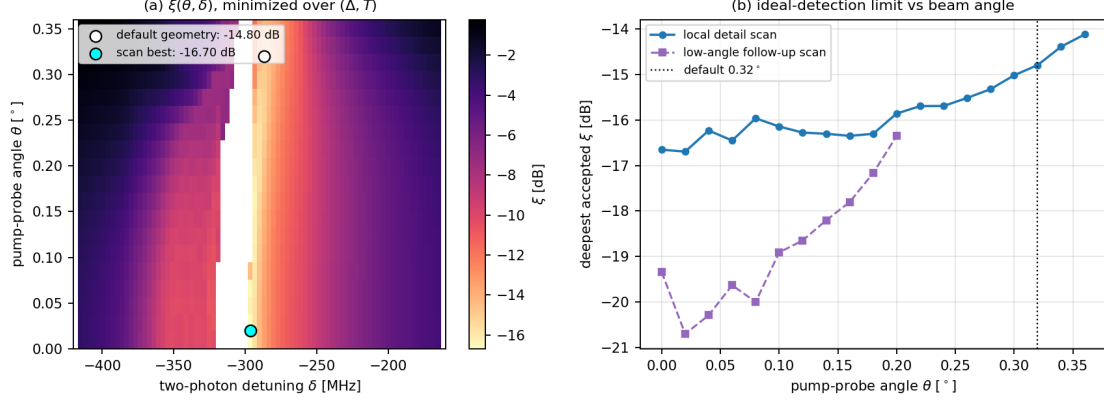
**Figure 7:** Ideal-detection ( $\eta = 1$ ) source limit of the final model. (a)  $\xi(\Delta, T)$ ; the optimum (circle) coincides in position with the finite-detection optimum of Fig. 6. (b) Deepest trusted squeezing at each temperature: with the excess-noise channels the optimum is capped at  $-15.6$  dB near  $110^\circ\text{C}$  (circle), whereas the cost-blind objective (dashed) runs away toward ever deeper values at high temperature. (c) The same comparison along the optimal-detuning slice.

### 7.4 Two-photon detuning and beam-angle geometry

The survey scans treat  $\delta$  as an internal optimization variable on a coarse grid and pin the pump-probe angle at the experimental  $0.32^\circ$ . A dedicated four-dimensional scan around the ideal-detection optimum ( $\Delta \in [-1.56, -1.44]$  GHz,  $T \in [96, 114]^\circ\text{C}$ ,  $\theta \in [0, 0.36]^\circ$ ,  $\delta \in [-420, -160]$  MHz, trust gate applied throughout) treats both explicitly (Fig. 8). At the default geometry the detailed scan finds its best trusted point at  $-14.8$  dB ( $\Delta = -1.54$  GHz,  $T = 106^\circ\text{C}$ ,  $\delta = -287$  MHz), compared with  $-15.6$  dB on the coarse survey grid (the two scans differ in  $\delta$  resolution and scan volume); in round numbers, the source limit of the default  $0.32^\circ$  geometry is  $-15$  dB. Relaxing the angle opens a distinct *low-angle ridge*: as  $\theta \rightarrow 0$  the longitudinal phase mismatch and the Gaussian-overlap penalty relax together, the optimal temperature falls, and the deepest trusted point reaches  $-20.7$  dB at  $\theta = 0.02^\circ$  ( $\Delta = -1.66$  GHz,  $T = 102^\circ\text{C}$ ,  $\delta = -307$  MHz). We deliberately report the ridge as a *geometry-conditional diagnostic*, not as a replacement optimum: in a real seeded FWM setup the crossing angle is tied to beam separation, spatial filtering, and pump-leakage rejection, and the pump-scatter coefficient of the model has not been recalibrated for near-collinear geometry. (The unconstrained scan also finds nominally deeper raw points, e.g.  $-27.6$  dB, but they fail the gain-gap gate,  $G_s - G_c \approx 0$ , and are rejected — the Stage-III guardrail doing its job.)

## 8 Tolerance of the recommended operating point

For experimental use, the location of the optimum matters less than how fast the squeezing degrades when a knob drifts. We therefore performed one-dimensional tolerance scans of the five experimentally accessible variables around the final optimum  $(\Delta, T, \delta) = (-1.50$  GHz,  $110^\circ\text{C}$ ,  $-280$  MHz), with everything else pinned (Fig. 9, Table 4). Each recomputed



**Figure 8:** Geometry study at ideal detection (trust gate applied; blank columns in (a) contain no trusted points — there the gain gap collapses and the noise formula is not meaningful, cf. Eq. (9)). (a)  $\xi(\theta, \delta)$  minimized over the local  $(\Delta, T)$  block; the white marker is the best point at the default  $0.32^\circ$  geometry, the cyan marker the best trusted point of the block. (b) Deepest trusted squeezing versus crossing angle, from the local detail scan and from a dedicated low-angle follow-up scan: the low-angle ridge reaches  $-20.7$  dB at  $\theta = 0.02^\circ$  but is tied to geometry calibration outside the model’s validated range.

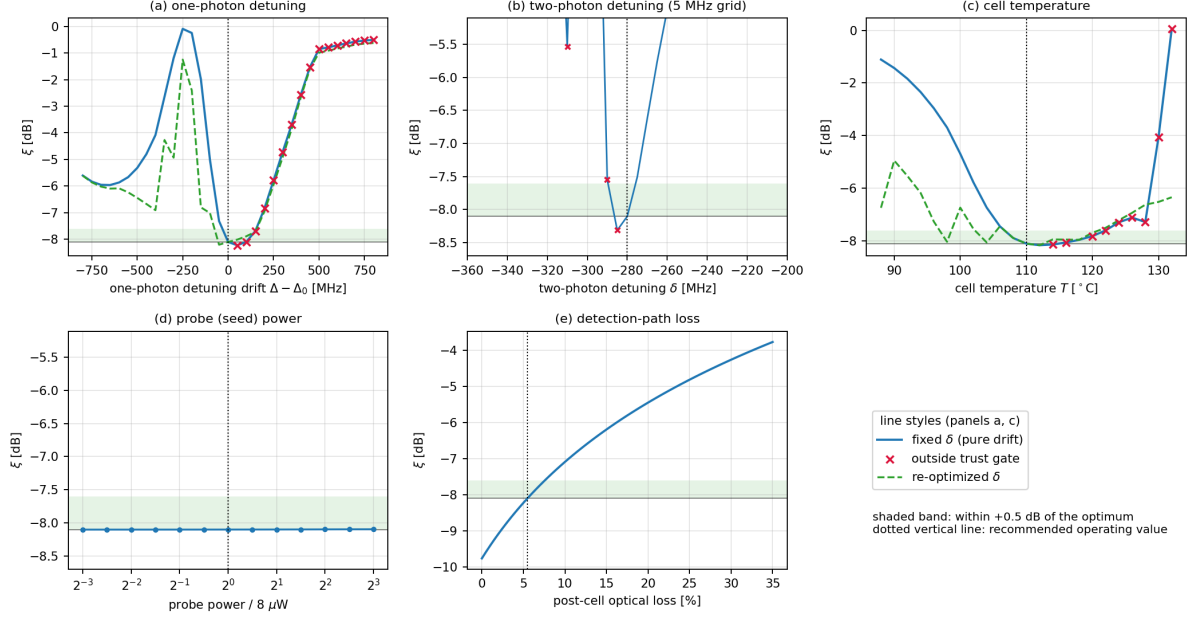
scan is read out in two ways: **fixed**  $\delta$ , modeling pure drift in the common configuration where the probe is derived from the pump by a fixed RF offset; and **re-optimized**  $\delta$ , modeling an experimenter who rescans the two-photon detuning after the drift. The detection-loss axis is an analytic remap of the stored gains (Sec. 2.2), verified to reproduce the solved value at the reference loss to machine precision. Degradation thresholds of  $+0.25$ ,  $+0.5$ , and  $+1.0$  dB correspond to  $+5.9\%$ ,  $+12.2\%$ , and  $+25.9\%$  in linear noise power; the discussion below uses  $+0.5$  dB as the representative budget.

One technical caveat uncovered here feeds back into the survey scans: the  $17.5$  MHz  $\delta$ -grid used in the wide scans can miss the narrow trusted minimum near a gate boundary and make a best- $\delta$  readout look worse than it is. The tolerance scans therefore use a  $5$  MHz grid; the optimum quoted in Sec. 7 is stable against this refinement.

**Table 4:** Tolerance windows around the recommended operating point (fixed- $\delta$  readout; drift measured from the optimum). The bracketed entries give the  $+0.5$  dB window when  $\delta$  is re-optimized after the drift.

| Variable        | +0.25 dB                                                | +0.5 dB      | +1.0 dB      | Remark                                                                                                                                                               |
|-----------------|---------------------------------------------------------|--------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $\Delta$ [MHz]  | $-16/+130$                                              | $-32/+155$   | $-55/+185$   | blue side enters the gap-collapse region; re-optimizing $\delta$ widens the $+0.5$ dB window to $[-75/+158]$ and, jointly with $\delta$ , to the $\pm 0.1$ GHz scale |
| $\delta$ [MHz]  | $-8.0/+2.1$                                             | $-9.6/+4.2$  | $-10.4/+7.3$ | narrowest absolute window; steep cliff on the blue side, spurious gate-failing dip near $-307$ MHz on the red side                                                   |
| $T$ [°C]        | $-2.2/+9.5$                                             | $-3.3/+12.0$ | $-5.0/+18.1$ | cliff on the cold side; the 1-D slice minimum sits near $112^\circ\text{C}$ ( $-8.17$ dB), so running $1\text{--}2^\circ\text{C}$ hot is the safe bias               |
| Seed power      | $1\text{--}64\ \mu\text{W}$ : no change ( $< 0.004$ dB) |              |              | model contains no seed-power-dependent technical noise; avoid detector saturation                                                                                    |
| Loss [%-points] | $+1.0$                                                  | $+2.1$       | $+4.5$       | slope $\simeq 0.22$ dB per %-point; eliminating the reference $5.5\%$ loss would deepen the optimum to $-9.77$ dB ( $1.67$ dB gain)                                  |

The practical checklist, in the order a lab should work through it when the squeezing is shallower than expected:



**Figure 9:** One-dimensional tolerance scans around the recommended operating point ( $\eta = 0.8694$ ). Solid: fixed  $\delta$  (pure drift); dashed:  $\delta$  re-optimized within the trust gate; crosses: fixed- $\delta$  points that leave the trust gate (their apparent depth is not to be believed — notably the seemingly favorable blue-side excursions in (a) and the dip near  $\delta = -307$  MHz in (b)). The shaded band is the +0.5 dB budget.

1. **Detection loss.** The only variable that moves the attainable floor itself: 0.22 dB is lost per percentage point, uniformly and irrecoverably. Check window transmissions, detector quantum efficiency, mode matching, and clipping first — and note that loss improvement is also the largest available upgrade (1.67 dB from removing the reference 5.5%).
2. **Two-photon detuning  $\delta$ .** The narrowest window ( $-9.6/+4.2$  MHz at +0.5 dB) with a cliff on the blue side — but it is set by RF electronics, so genuine drift is unlikely; the real risk is targeting the wrong value. It is also the cheapest knob to verify (an RF rescan), so rescan  $\delta$  immediately after checking loss. Do not let an optimizer walk into the seemingly deep dip near  $-307$  MHz: it fails the gain-gap gate and is an artifact.
3. **One-photon detuning  $\Delta$ .** The effective red-side window is about 32 MHz; apparently favorable blue-side values sit in the gap-collapse region and should not be trusted. A pump lock at the 10 MHz level is appropriate. Because  $\Delta$  and  $\delta$  are strongly coupled, always rescan  $\delta$  after moving  $\Delta$  — jointly they tolerate an order of magnitude more drift.
4. **Temperature.** Asymmetrically forgiving ( $-3.3/+12^\circ\text{C}$ ); controller stability is a non-issue and the practical risk is absolute calibration (sensor offset). Bias  $1\text{--}2^\circ\text{C}$  hot, away from the cold-side cliff.
5. **Seed power.** Insensitive across  $1\text{--}64\mu\text{W}$  in the model (the susceptibilities are seed-independent and pump depletion is negligible at these powers); experimentally, only detector saturation matters. Check last.

## 9 Discussion

### 9.1 What limits the squeezing — and what does not

Across all five stages one hierarchy has remained stable. The *depth* of the observable squeezing is set almost entirely by the detection efficiency: once  $G_s \gtrsim 30$ , additional gain buys nothing (Fig. 1), while every percentage point of detection loss costs  $\approx 0.2$  dB irrecoverably. The *location* of the optimum, by contrast, is set by in-cell absorption — the competition between parametric



gain and the linear optical depths of *both* sidebands — as demonstrated by the fact that removing the detection floor entirely does not move the optimum (Sec. 7). Finally, loss of any kind can only push the intensity-difference noise toward shot noise, never above it (Eq. (3)); measured anti-squeezing is always excess noise, with the leakage of the gain-amplified intensity-sum noise through detector or loss imbalance the most dangerous channel at high gain. This is one more reason the efficient-frontier philosophy — reach the floor at the lowest temperature, gain, and optical depth — is also the robust experimental choice.

## 9.2 A guardrail checklist for simulation-based squeezing predictions

Each of the defects in this study was caught not by comparison with experiment but by an internal-consistency check. We collect them here as a checklist for similar simulation efforts:

1. **Detection-floor bound.** At detection efficiency  $\eta$ , no point may be deeper than  $10 \log_{10}(1 - \eta)$  (Eq. (4)). This holds regardless of the source model; treat any violation as a numerical error, and treat a large *plateau at the floor* as a sign that the optimization question itself is degenerate (Stage I).
2. **Conserved-quantity consistency gate.** Enforce the physical constraint of the process — here  $G_s - G_c = 1$  for lossless seeded parametric gain, gated as Eq. (9) — and exclude untrusted points from optimization. A formula that is exact on the constraint manifold can be spectacularly wrong off it (Stages III and V: background points and raw low-angle optima reporting spurious perfect squeezing).
3. **Formula cross-check on identical data.** When two expressions for the same observable exist, evaluate both on the same stored intermediate data. A divergence that *grows with gain* indicates a wrong asymptote; sensitivity to floating-point cancellation indicates an expression that must be reformulated in terms of small differences (Stage III).
4. **Cost-charging test.** Ask whether the optimum survives when the physical costs of its own figure of merit are charged. An optimum whose reported diagnostics are internally contradictory — here,  $G_s = 636$  with zero reported absorption — is an artifact of an objective blind to a channel (Stage IV).
5. **Sign and convention audit.** Write the frequency/sign relations down symbolically and verify every code path against them; small-parameter sign errors distort rather than destroy the landscape and can survive long scrutiny (Stage V). Re-anchor all regression baselines after each physics fix.
6. **Resolution audit at gate boundaries.** A coarse grid in the optimized variable can interact with a trust gate to misreport the best trusted value (the 17.5 MHz versus 5 MHz  $\delta$ -grid in Sec. 8).

## 9.3 Limitations

The model is a steady-state description with a fixed thermal ground-state population; it does not include optical pumping, which is the missing physics most likely to decide the red/blue near-degeneracy of Sec. 6 in favor of the conventional blue-side operating point. The pump-scatter coefficient is calibrated only at the reference geometry, so the low-angle ridge of Sec. 7 is a design target, not a prediction. Seed-power-dependent technical noise (detector saturation, laser intensity noise) is outside the model, as are all excess-noise channels beyond per-sideband absorption and pump scatter; the quoted depths are therefore best-case values for the efficiency-and-absorption budget alone, not guaranteed measurement outcomes. Finally, the corrected noise formula is itself limited by double-precision arithmetic above  $G \sim 10^7$ , far outside the recommended operating regime.

## 10 Conclusion

A first-principles steady-state simulation of double- $\Lambda$  four-wave mixing on the  $^{85}\text{Rb}$  D1 line, calibrated once against the reference experiment of Ref. [4], predicts a recommended operating point at  $\Delta = -1.50$  GHz,  $T = 110^\circ\text{C}$ ,  $\delta = -280$  MHz with  $\xi = -8.10$  dB at  $\eta = 0.8694$  — within 0.7 dB of the detection floor at a moderate gain of  $G_s \approx 23$  — and reproduces the measured squeezing, gain, and temperature scales of the reference experiment at its own operating point. The tolerance budget around the recommended point is dominated by detection loss (0.22 dB per percentage point) and by the two-photon detuning ( $-9.6/+4.2$  MHz for  $+0.5$  dB), with temperature asymmetrically forgiving and seed power irrelevant; when squeezing is lost, the efficient checking order is loss, then a  $\delta$  rescan, then the pump lock, then temperature calibration, then seed power. At ideal detection the same operating point reaches  $-15.6$  dB, and a near-collinear geometry could in principle reach  $-20.7$  dB, though the latter requires geometry calibration beyond the model’s validated range. The red-side optimum is nearly degenerate with the conventional blue-side operating point (0.32 dB), and deciding between them requires optical-pumping physics absent from the present model.

Equally important are the negative results: a  $-82$  dB “theoretical optimum” that was the artifact of a catastrophically canceling formula, an extreme-gain optimum produced by an objective function that failed to charge for absorption, and a sign error that survived four model revisions. All were caught by cheap internal-consistency guardrails — a detection-floor bound, a conserved-quantity gate, formula cross-evaluation on identical data, and a symbolic sign audit — that we recommend as standard practice for any simulation used to steer a squeezing experiment.

## Data availability

The complete simulation source code, the archived scan data underlying every figure, the figure-generation scripts, and the regression tests that pin the calibrated reference behavior are openly available in the GitHub repository of Ref. [6]. The historical scans of Stages I–II are shown in their archived rendered form (Figs. 2 and 3), as their raw grids were later overwritten; all other figures are regenerated directly from the archived raw data.

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